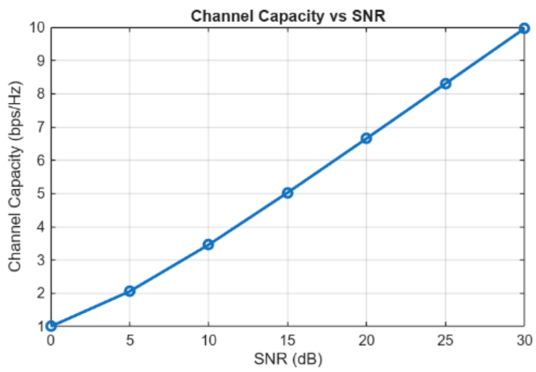
OBJECTIVE 1:

```
clc; clear;

close all;

snr = 0:5:30;

snr_linear = 10.^(snr/10);

capacity = log2(1 + snr_linear);

figure; plot(snr,capacity,'-o','LineWidth',2,'MarkerSize',6);

grid on;

xlabel('SNR (dB)');

ylabel('Channel Capacity (bps/Hz)');

title('Channel Capacity vs SNR');
```

OUTPUT :

```
clc;
clear;
close all;
snr = 0:5:30;
snr_linear = 10.^(snr/10);
capacity = log2(1 + snr_linear);
figure;
plot(snr,capacity,'-o','LineWidth',2,'MarkerSize',6);
grid on;
xlabel('SNR (dB)');
ylabel('Channel Capacity (bps/Hz)');
title('Channel Capacity vs SNR');
```

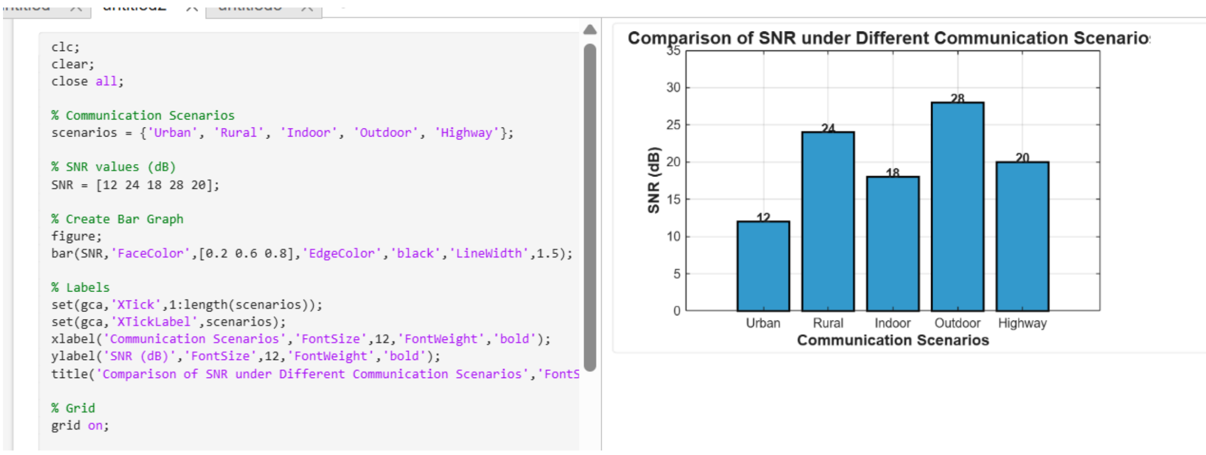


OBJECTIVE 2:

```
clc;

clear;

close all;

% Communication Scenarios

scenarios = {'Urban', 'Rural', 'Indoor', 'Outdoor', 'Highway'};

% SNR values (dB)

SNR = [12 24 18 28 20];

% Create Bar Graph

figure;

bar(SNR,'FaceColor',[0.2 0.6 0.8],'EdgeColor','black','LineWidth',1.5);
```

```
% Labels

set(gca,'XTick',1:length(scenarios));

set(gca,'XTickLabel',scenarios);

xlabel('Communication Scenarios','FontSize',12,'FontWeight','bold');

ylabel('SNR (dB)','FontSize',12,'FontWeight','bold');

title('Comparison of SNR under Different Communication
Scenarios','FontSize',14,'FontWeight','bold');

% Grid

grid on;

% Display SNR values on top of bars

for i = 1:length(SNR)

    text(i, SNR(i)+0.5, num2str(SNR(i)), ...

        'HorizontalAlignment','center', ...

        'FontWeight','bold');

end

ylim([0 35]);
```

OUTPUT:



OBJECTIVE 3:

```
clc;

clear;

close all;

antennas = [4 8 16 32 64]; % Number of Antenna Elements
```

gain = 10*log10(antennas); % Antenna Gain (dB)

figure

bar(antennas, gain)

grid on

xlabel('Number of Antenna Elements')

ylabel('Antenna Gain (dB)')

title('Antenna Gain for Different Antenna Array Sizes')

OUTPUT: